The Size-Height Correlation: An argument for cyclic prosodification

Paula Fenger, Nadja Fiebig, Sören Tebay & Philipp Weisser

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Introduction

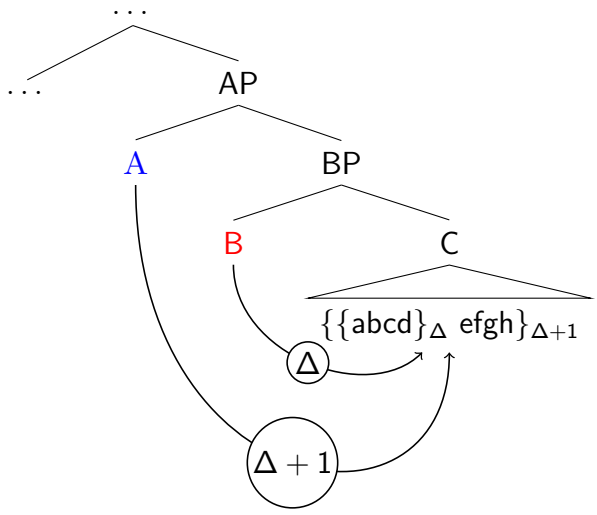
- We provide two crosslinguistically robust generalizations concerning the Morphosyntax-Phonology Interface coming from independent phenomena:
  - Verbal Number Reduplication
  - Prosodically determined second position coordinators
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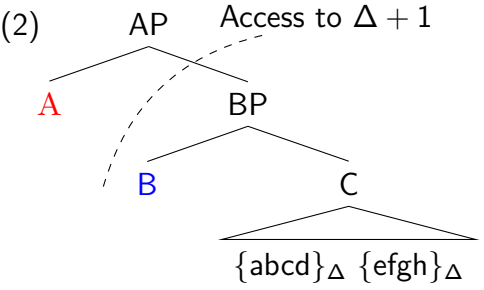
(1) The Size-Height-Correlation:



- We claim that this provides a strong argument for a cyclic account of the prosodification of syntactic structure.

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- At the point where B is inserted and triggers the application of a phonological process, only the lower prosodic domain Δ is built
  - ↪ The higher domain $\Delta + 1$ is not accessible yet as it is only completed after B.

(2)



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Take-home message:

- The Size-Height Correlation indicates that smaller domains must be construed and become accessible before bigger domains.
- This finding is at odds with parallel computation of prosodic domains (Truckenbrodt 1999; Selkirk 2011; Elfner 2015 and many others).

Flavours of Verbal Number

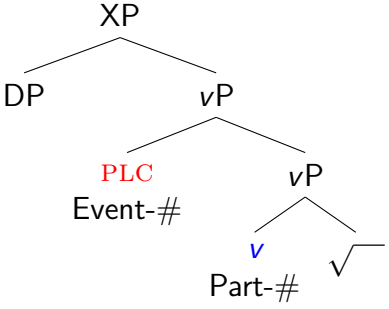
- The morphosyntactic category of VERBAL NUMBER comes in two flavours:
  - Event Number (as in: 'to pet the dog several times')
  - Participant Number ('as in: 'to pet several dogs')

(3) ko-roːp-ɑn∼ɑ∼roːp.ɑn-i
PST-rain-VBLZ∼LK∼RED-IPFV
'It rained a lot/frequently.'

Kipsigis, event number (Fiebig field notes)

- Following Thornton (2019); Amato (2018), Fiebig (2024) argues that participant number is a head that is very close to the verb root (either it is on *v* or below *v*).
- Event number is higher (above vP).

(4)



- Both types of verbal number are frequently expressed by means of reduplication.
- Crucially, we find a correlation between the size of the reduplicant and the type of verbal number

(5) a. kua ha-hala e ia e tau lā akau.
PERF RED.PART-cut ERG he ABS PL branch tree
'He cut the branches.'

b. ne noko-noko e ia e gutuhala
PST RED.EVENT-knock ABS she ABS door
'She knocked on the door (many times).'

Niuean, Haji-Abdolhosseini et al. (2002)

Verbal number and reduplication: Typology

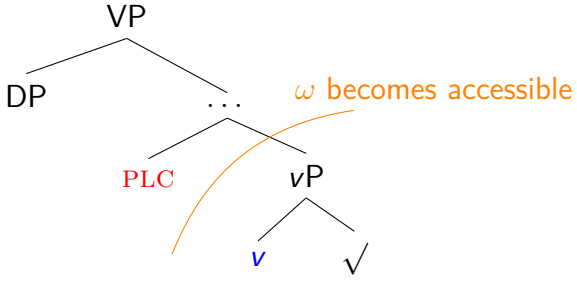
- Crosslinguistically, partial reduplication marks both flavours of verbal number, but full reduplication marks only event number. (Wood 2007, Mattiola 2019, Bybee et al. 1994)

language	partial reduplication	full reduplication
Acoma	participant number	event number
Amele	-	event number
Burmese	-	event number
Daga	event number	-
Hausa	participant number	-
Jarawara	participant number	-
Kipsigis	-	event number
Papanui	participant number	event number
Sonora Yaqui	participant / event number	-
Wari	participant number	event number
Pero	participant number	-
Yurok	participant number	-

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- The higher head can target both the ω-level and or a smaller constituent.
- This can be made sense of by assuming that the crucial ω is only accessible after the vP is completed.

(6)

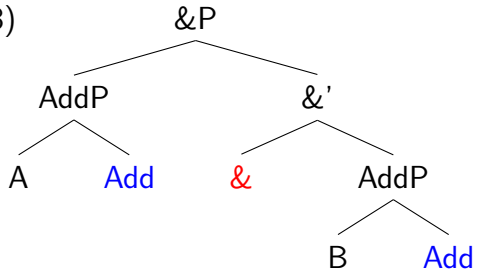


Shifting Coordinators

- Languages typically use two morphological strategies to coordinate elements.

(7)
a. Petyr-en Maša-jen
Peter-ADD Masha-ADD
b. Petyr no Maša
Peter and Masha
'Peter and Masha'
c. Petyr-en no Maša-jen
Peter-ADD and Masha-ADD
'Peter and Masha'
Udmurt, (Weisser, 2017)

(8)



e.g. Bhatt (2014),
Mitrović and Sauerland (2014)

- Some clause-level coordinators do not appear in their canonical position (e.g. in between the two clauses) but shift to a second position inside one of their coordinands.
  - ↪ And in some of the cases, this position is prosodically defined:

(9) [Is istum reliqui-t] [{de provincia}$_{\omega}$=que decess-it]
he it leave.PERF-3SG from province=AND depart.PERF-3SG
'He left it and departed from the province.'
Cicero, Against Verres 2.2.48

(10) ... {Ọlá yóò}$_{\phi}$ sì lọ.
... Ola will AND go
'...and Ola will go.' Yorùbá, Aremu and Weisser (2024)

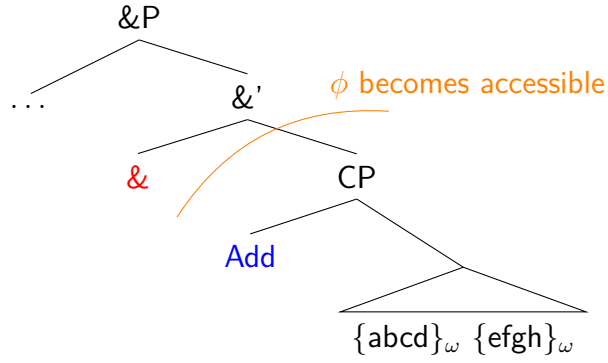
- In Weisser (2024), I show that we find a bidirectional correlation between the syntactic type of the coordinator (& vs ADD) and the prosodic constituent that it shifts across in order to get to a 2nd position.

(11) Phonol. defined position and syntactic status:

	[ω-COORD	[ϕ-COORD
ADD	Latin Cherokee Kalaallisut Anc. Greek Hittite Gothic Old Irish	–
&	–	Yorùbá German Polish

- Again, we observe a correlation between the syntactic height of the coordinator and the size of the prosodic constituent that is affected by it.
- Only once the CP is built, the ϕ-phrases within it become an accessible target.

(12)



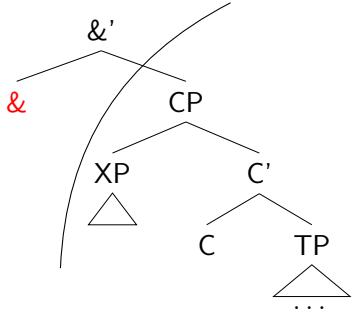
- In Weisser (2024), I sketch an approach that derives the pattern in a cyclic framework, in which linearization statements as well as prosodification are generated as part of a bottom-up Vocabulary Insertion process.
  - ↪ As soon as VI hits a syntactic node in the structure, it will map it into a prosodic constituent and linearize it wrt. the previously generated structure. (cf. Kalin 2022)

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- Second position placement arises as a subcategorization-driven transformation (INTEGRATION) that dislocates an element X to a position after the first element within the category it precedes.

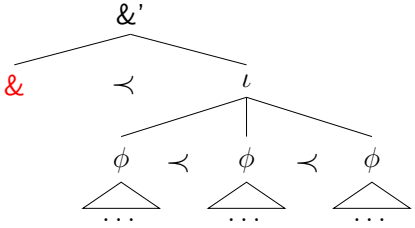
Phonological 2nd positions - A monosyndetic coordinator

- The configuration at Spell-Out (13) the configuration at Vocabulary Insertion (14)

(13)



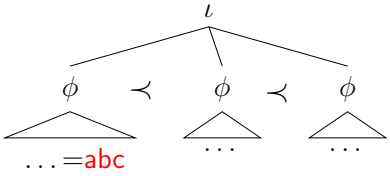
(14)



(15) [&] ⟷ /-abc/ [➪➪]

The result of INTEGRATION:

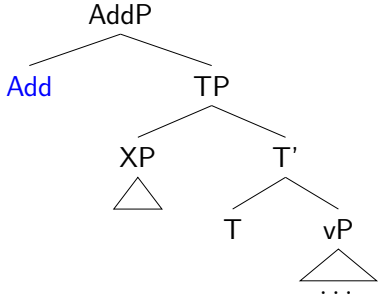
(16)



Phonological 2nd positions - A polysyndetic coordinator

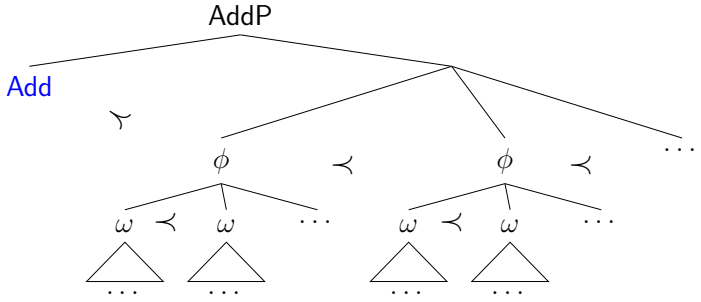
- The configuration at Spell-Out (17)

(17)



- The configuration at Vocabulary Insertion:

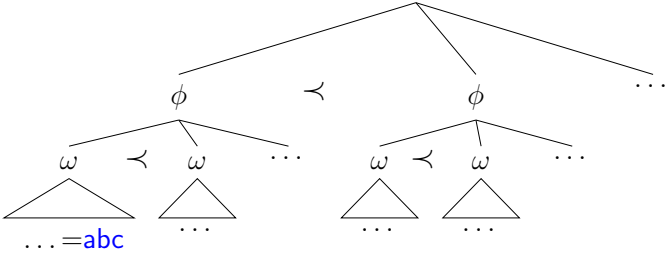
(18)



(19) [Add] $\longleftrightarrow$ /-abc/ [⇨⇦]

The result after INTEGRATION

(20)



- The basic building blocks (e.g. Vocabulary Insertion, Prosodification, Integration) all work the same in both configurations.
- The difference in shifting behavior of the two types of syntactic elements arises purely as the result of the different timing.
  - ↪ It is crucial that prosodification applies cyclically.
  - ↪ As soon as a given cyclic node in the tree is reached (e.g. CP), it will be mapped to the respective prosodic category.
- Only when there is a CP, the complement of the coordinator will be mapped to a ι-phrase, thereby changing target of the INTEGRATION operation.

Conclusion:

- We reported two case studies of robust crosslinguistic generalizations about the respective subtypes of a morphosyntactic category and the size of the prosodic domain they affect if they trigger a phonological process.

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- We have argued that the Size-Height Correlation presents a strong argument for a cyclic conception of prosodification.

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